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Ambiguity-Free Geometrical Deep Space Navigation Using Angle-Only Measurements

Background

Many deep space missions rely on the Deep Space Network (DSN) for navigation. However, for CubeSat missions, access to the DSN may be limited due to its high operational costs. As a result, recent research has focused on developing onboard navigation systems that are computationally efficient. These onboard solutions not only reduce reliance on DSN but also support greater autonomy by eliminating delays associated with signal transmission.

In this study, we present a geometric approach to deep space navigation using angle-only measurements [1]. We assume that the positions of the Sun, Planet 1, and Planet 2 are known a priori from ephemeris data. The spacecraft then measures the angles formed between itself, the Sun, and each of the two planets, specifically, the angles Sun–spacecraft–Planet 1 and Sun–spacecraft–Planet 2. Using this information, the proposed method can compute the spacecraft's distance from the Sun. While this approach introduces sign ambiguities in some angles during the process, resulting in up to nine possible solutions, we also demonstrate how these ambiguities can be effectively resolved.

Once the distance from the Sun to the spacecraft is determined, the spacecraft's inertial position can be computed [2]. At this stage, a sign ambiguity arises again when expressing the spacecraft's position in a reference frame. Although this is a geometric limitation [2], the ambiguity can be resolved using prior information, such as a known reference orbit or estimation results from the previous step. After determining the inertial position, the spacecraft's attitude relative to the inertial frame can be calculated using the TRIAD method. Thus, the proposed approach enables determination of both the spacecraft's inertial position and its attitude.

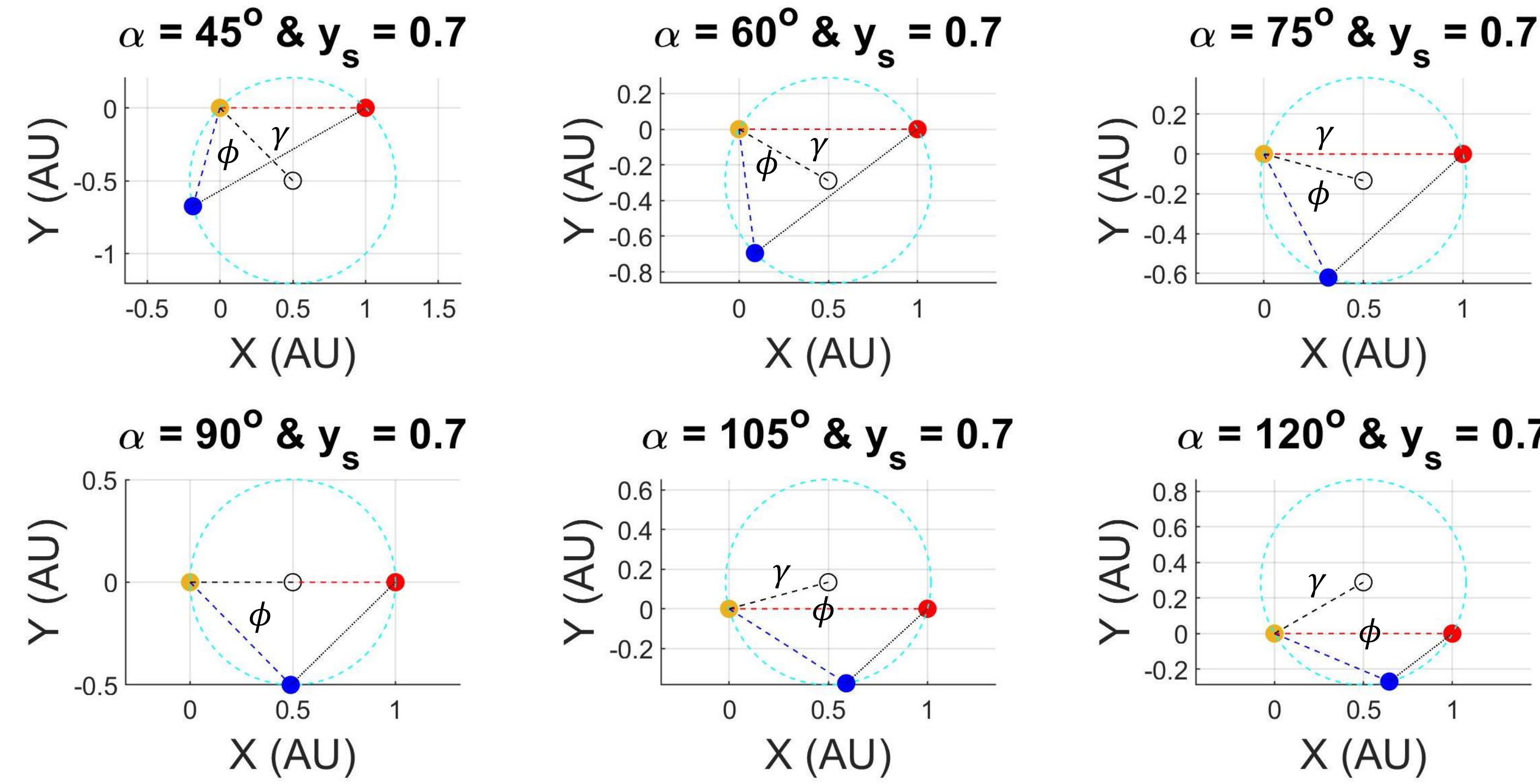
Geometry

The initial geometry is shown in the bottom-left figure. The Sun (orange) is located at the origin, and the spacecraft (S/C) is represented by a black star positioned along the Y-axis. Planet 1 and Planet 2 are shown in red and blue, respectively. The $\vec{r}_1$ and $\vec{r}_2$ represent the positions of the planets from the Sun, while r_3 denotes the distance between Planet 1 and Planet 2. They are given from ephemerides. The unit vectors $\hat{\rho}_0$, $\hat{\rho}_1$, and $\hat{\rho}_2$ correspond to the angle-only measurements taken by the spacecraft, which allow us to compute the angles α_1 and α_2 . The angle θ represents the angular separation between Planet 1, the Sun, and Planet 2, computed from the dot product of $\vec{r}_1$ and $\vec{r}_2$. The distances from the spacecraft to Planet 1 and Planet 2 are denoted as y_1 and y_2 , respectively.

To compute the distance from the Sun to S/C (y_s), a circumcircle is constructed, as shown in the bottom-middle figure. C_1 and C_2 represent the centers of the circumcircles formed by the Sun, the spacecraft, and Planet 1 and Planet 2, respectively. The radii of these circumcircles, R_1 and R_2 , are calculated using the Law of Sines. The angles γ (Planet–Sun– C_1) and ϕ (C_1 –Sun–S/C) can be determined through simple trigonometric relationships. For further analysis, spherical geometry is introduced, as illustrated in the bottom-right figure. The angle between the spacecraft, the Sun, and a planet (S/C–Sun–Planet) is denoted by δ , and it is defined as: $\delta = s_\phi \phi + s_\gamma \gamma$, where s_ϕ and s_γ are the sign of ϕ and γ , respectively, capturing the inherent sign ambiguities. Within this unit sphere framework, the spherical law of cosines can be applied. Because the only unknown in the previous equation is y_s , it can be computed by solving the root-finding problem. Since δ depends on sign-ambiguous angles, up to nine possible solutions may arise.

Sign Ambiguity: s_γ

The first sign ambiguity for γ can be resolved in a straightforward manner. The figure below illustrates how the geometry changes with fixed y_s as the angle α varies. As α increases from 45° to 75° , the angle γ decreases. During this process, the total angle $\delta = \phi + \gamma$, representing the angle between S/C (blue), the Sun (orange), and the Planet (red). When α reaches 90° , γ becomes zero. As α continues to increase beyond 90° , to 105° and 120° by shaping the obtuse triangle, the relationship changes to $\delta = \phi - \gamma$. Therefore, from this observation, when α is smaller than 90° , then the sign of γ is positive ($s_\gamma = +1$), and when α is greater than 90° , the sign becomes negative ($s_\gamma = -1$).



Test Case

A test case is presented in the upper table. The indices 1, 2, and 3 refer to the subscripts of each parameter. P_x , P_y , and P_z represent the x-, y-, and z-components of the position vector, respectively. In this case, Mars and Jupiter are selected as the two reference planets. The variable r denotes distance, while R refers to the radii of the circumcircles. Ultimately, the computed value of y_s is 1.4323 AU, which exactly matches the true distance.

The lower table presents the relevant parameters under different combinations of signs for $s_\phi^1, s_\phi^2, s_\gamma^1$ and s_γ^2 , shown from left to right. Among all combinations, the one yields the smallest determinant value D , and is therefore identified as the correct solution

Parameters	1	2	3
P_x (AU)	-1.3484	4.5648	0 (S/C)
P_y (AU)	0.9816	1.9726	-1.3423 (S/C)
P_z (AU)	0.0536	-0.1103	0 (S/C)
r (AU)	1.6687	4.974	5.9979
α (°)	30.143	54.021	84.162
R (AU)	1.6615	3.0733	N/A
ϕ (°)	60.175	77.386	N/A
γ (°)	59.857	35.979	N/A
y (AU)	2.6873	5.6425	1.4323 (y_s)

$s_\phi^1 s_\phi^2 s_\gamma^1 s_\gamma^2$	$-,-,+,+$	$+, -, +, +$	$-, +, -, +$	$+, +, -, +$
δ_1 (°)	-6.3187	126.03	-6.3187	126.03
δ_2 (°)	-41.407	113.36	-41.407	113.36
y_1 (AU)	-0.3657	2.6873	-0.3657	2.6873
y_2 (AU)	-4.0654	-4.0654	5.6425	5.6425
D	-19.616	-10.004	-3.5832	-3.55e-14

Sign Ambiguity: s_ϕ

The second sign ambiguity, associated with ϕ . This ambiguity arises from the geometric configuration relative to each planet, as illustrated in the figure on the right. In this setup, the Sun (orange) is positioned at the origin, and Planet 1 (P1), marked in red, lies on the X-axis, 1 AU from the Sun. S/C is located on the blue and green circles, corresponding to a distance $y_s = 1.0816$ AU, referred to as *Case a* and *Case b*. In this configuration, the measured angles α^a and α^b are identical, as they subtend the same arc between the Sun and Planet 1 on the circumcircle. However, the angles ϕ^a and ϕ^b , shown in blue and green respectively, differ only in sign. That is, the difference lies in the sign of ϕ , denoted by s_{ϕ^a} and s_{ϕ^b} . In *Case a*, the total angle is given by $\delta = \gamma + \phi$, while in *Case b*, it becomes $\delta = \gamma - \phi$.

To resolve the sign ambiguity of s_ϕ , examining a single geometry is not sufficient. Instead, we utilize measurements from both planets and refer back to the initial configuration. The angle between Planet 1–spacecraft–Planet 2, denoted α_3 , can be directly computed from the measurements. Once δ is known, the distances y_1 and y_2 (from the spacecraft to each planet) can be determined using the Law of Sines. We define a determinant function D as the absolute error from the Law of Cosine using the triangle with y_1, y_2 , and r_3 . For each planetary geometry, up to two possible values of δ may exist, leading to four total combinations of sign choices. Among these, we select the combination that yields the smallest value of D . Through this iterative process, the correct signs of ϕ can be determined at each step.

Conclusion

The entire process is summarized in the right flow chart. Starting with inputs (ephemerides, measurements, and an initial guess $\hat{y}_s$), we compute the intermediate parameters. When the measurement angle α is greater than 90° , we set $s_\gamma = -1$ and $s_\phi = +1$; otherwise, $s_\gamma = 1$. If sign ambiguities still remain, we evaluate the determinant (D) and select the sign combination that results in the smallest D . Finally, we solve the root-finding problem iteratively, updating $\hat{y}_s$ until convergence is achieved.

In our test case, the proposed method demonstrates robustness with respect to the initial guess of y_s . Typically, a reliable initial guess can be derived from prior information, such as a reference trajectory. However, even when a poor initial guess was used, with an error of about 10% in y_s , the method was still able to successfully converge to the correct solution without any issues.

As the next step, we aim to evaluate the performance of the proposed method under measurement uncertainties. In practice, sensors introduce errors when measuring the unit vectors of planets or the Sun. To assess the impact of these uncertainties, we plan to conduct a Monte Carlo simulation and analyze the resulting accuracy of y_s . This resulted y_s error can later be mitigated through filtering techniques within the same project framework.

References

- [1] Negrete, Aimar, and Ossama Abdelkhalik. "A geometric approach to deep space navigation using angles-only measurements." *AIAA SCITECH 2024 Forum*. 2024.
- [2] Fang, Bertrand T. "Trilateration and extension to global positioning system navigation." *Journal of Guidance, Control, and Dynamics* 9.6 (1986): 715-717.

